Cameron Severn

SENIOR SOFTWARE ENGINEER | PLATFORM, ML & AI

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SUMMARY

Scientist and technologist with an MS in Biostatistics who progressed to Principal Data Scientist by combining statistical rigor, classical ML, and hands-on software engineering. Applied regression, survival analysis, XGBoost, experimental design, and explainable ML across biomedical and telecom problems, then carried that rigor into production service ownership, AI platforms, and agentic engineering. A wide-ranging path through biology, audio engineering, data science, and software reflects curiosity and adaptability: synthesizing unfamiliar information, ramping quickly across languages and domains, and translating difficult technical ideas into plain language people can act on.

EXPERIENCE

Principal Data Scientist (promoted from Data Scientist VI and IV) | Charter Communications

Greenwood Village and Denver, CO | Dec 2021 - May 2026

- Owned a production network-quality monitoring service with an engineering team: built and shipped ML models and batch pipelines, kept recurring jobs reliable, shared pager/on-call coverage, and improved the system while it remained live.
- Progressed from contract Data Scientist IV to Principal while building data, engineering, and machine-learning software at scale.
- Applied statistical inference, linear regression, XGBoost, and large-scale experimentation to network-health measurement, capacity planning, and proactive maintenance.
- Contributed to Charter systems using Keycloak-based authentication and SSO.
- Named co-inventor on two pending Charter U.S. patent applications covering ML-based impaired-connectivity identification and network-demand forecasting.
- Translated complex statistical and engineering concepts for technical and nontechnical partners while mentoring junior data scientists and engineers.

Research Instructor & Research Assistant | University of Colorado

Aurora and Denver, CO | Aug 2017 - Dec 2021

- Delivered biostatistical consulting and analysis across Pediatrics - Endocrinology, the Cancer Center Biostatistics Shared Resource, and Biostatistics & Informatics.
- Used regression, survival methods, experimental design, explainable ML, and data visualization to turn complex biomedical data into defensible results; co-authored 21 peer-reviewed journal articles.
- Conducted molecular-biology research on Zygote Arrest proteins using PCR, gel electrophoresis, Western blots, protein/RNA extraction, cloning, and dual-luciferase assays; generated publication data and figures.

SELECTED PLATFORM & AGENTIC ENGINEERING

Wildcard Dex | Published AI Product

- Published and operated an AI-backed Flutter app on the App Store and Google Play using Firebase Auth, Cloud Functions, Firestore, and Gemini; hardened model flows with server-side guards, strict parsing, moderation, bounded retries, observability, and safe localized failures.

Sleep Mask | Published Adaptive Audio Product

- Published a privacy-first Flutter app on Google Play that synthesizes adaptive noise in real time, analyzes room sound on-device with FFT and Bark-band processing, and reshapes output without recording or transmitting microphone audio.

Slushnet | Agentic Engineering Platform - In Development

- Built hands-on agentic engineering workflows across Claude Code, Codex, coding agents, local runtimes, and API models using reusable skills, bounded loops, execution graphs, dynamic routing, tools, prompt/context pipelines, evals, and MCP integrations.
- Implemented a Rust/TypeScript harness with typed planner, writer, parallel-review, join, and approval stages, durable checkpoints, bounded fallback, and inspectable guardrails for credentials, spend, protected effects, evidence, and human gates.

SK8 | C#/.NET 8 Engineering System - In Development

- Built a modular C#/.NET 8 codebase with a deterministic engine-independent core, schema-driven persistence, replay compatibility, broad automated coverage, and a bounded AI-assisted development control plane.

STATISTICAL ML & RESEARCH

- Worked across statistical inference, experimental design, linear regression, discrete-time survival models, gradient-boosted trees (XGBoost), regression-based kernel methods, radiomics, explainability, and data visualization.
- Co-authored 21 peer-reviewed journal articles across statistical methods, medical imaging, computational biology, and clinical research; Google Scholar recorded 575 citations and an h-index of 12 in Aug 2026.

SELECTED PUBLICATIONS

- First author: A Pipeline for the Implementation and Visualization of Explainable Machine Learning for Medical Imaging Using Radiomics Features. Sensors 22(14), 5205 (2022).
- Survival Prediction Models: An Introduction to Discrete-time Modeling. BMC Medical Research Methodology 22, 207 (2022).
- PaIRKAT: A Pathway Integrated Regression-based Kernel Association Test With Applications to Metabolomics and COPD Phenotypes. PLoS Computational Biology 17(10), e1008986 (2021).
- Novel Clinical Algorithm for Hypothalamic Obesity in Youth With Brain Tumours and Factors Associated With Excess Weight Gain. Pediatric Obesity 17(7), e12903 (2022).
- Prediction of Resting Energy Expenditure for Adolescents With Severe Obesity: A Multi-centre Analysis. Pediatric Obesity 19(7), e13123 (2024).

INTELLECTUAL PROPERTY

- Network Planning Through Network-Demand Forecasting and Simulation - Co-inventor; published as US 2026/0197242 A1; pending.
- Co-inventor on an additional pending U.S. application involving ML-based network reliability.

SKILLS

Statistics & ML: Statistical inference, experimental design, linear regression, survival analysis, XGBoost, explainable ML, model evaluation, data visualization, R, Python

Agentic engineering: Claude Code, Codex, model APIs, LLM orchestration, skills, loops, execution graphs, dynamic workflows, agent/tool harnesses, prompt/context pipelines, evals, MCP, guardrails

Identity & security: Keycloak SSO, Clerk authentication and user identity, Firebase Auth, authenticated APIs, protected routes

Engineering & data: Python, TypeScript/JavaScript, Scala, C#/.NET 8 (recent project work), Rust, Node.js, Dart/Flutter, Firebase, Cloud Functions, Firestore, SQLite/Drift, AWS, CI/CD

EDUCATION & CERTIFICATION

MS, Biostatistics (Data Science Analytics) | Colorado School of Public Health

Thesis: Explainable Predictions of IDH Mutation in Glioma; built an explainable machine-learning pipeline using MRI radiomics features and interactive visualization to support transparent clinical decisions.

BS, Biology; BS, MEIS - Recording Arts | University of Colorado Denver

Additional training and certification: TensorFlow in Practice Specialization | Survival Analysis in R for Public Health | AWS Certified Cloud Practitioner